GulzarSoft

Employee Evaluation Task (AI-RAG)

Foreword

The following task has been designed primarily to evaluate the candidate's interest level, problem-solving ability, and AI engineering skills. Completing it within the due time is a plus. Maintain a work log explaining your approach.

Task

Build a Retrieval-Augmented Generation (RAG) system for a customer support team. The system should help support agents quickly retrieve accurate information from product documentation and historical support tickets to answer customer queries.

Time Limit: 3–4 hours
Difficulty: Intermediate (Suitable for Fresh AI Engineers)

Dataset Description

Three JSON files will be provided:
• product_docs.json – Product documentation (setup, troubleshooting, billing, security, API, etc.)
• support_tickets.json – Historical support tickets with resolutions.
• test_queries.json – Test queries for evaluation.

Core Requirements

Part 1 – Document Processing & Indexing (45 min)
• Load & parse all JSON documents.
• Intelligent chunking while preserving steps and metadata.
• Generate embeddings and store in a vector database.

Part 2 – Smart Retrieval (1 hour)
• Hybrid retrieval (semantic + keyword).
• Document prioritization.
• Query classification.
• Synonym & negation handling.
• Re-ranking by relevance and recency.

Part 3 – Response Generation (45 min)
• Generate answers with citations, confidence indicators, and proper formatting.
• Handle insufficient, conflicting, and outdated information gracefully.

Complexity Challenge (Choose One)

A. Version-Aware Information Management
B. Multi-Document Reasoning
C. Context-Aware Troubleshooting

Deliverables

1. Working RAG System
2. README, Architecture & API Documentation
3. Evaluation Report
4. Clean, well-commented code, logging, unit tests, and requirements.txt

Recommended Tech Stack

Python: scikit-learn, sentence-transformers, OpenAI/Gemini embeddings, LangChain, ChromaDB.
JavaScript: TensorFlow.js, LangChain, Pinecone, OpenAI/Gemini.

Deadline

The deadline for submitting your work log and source code will be shared separately via email.
Please document your assumptions and work log.

Submit your response to:
zainali.offical789@gmail.com